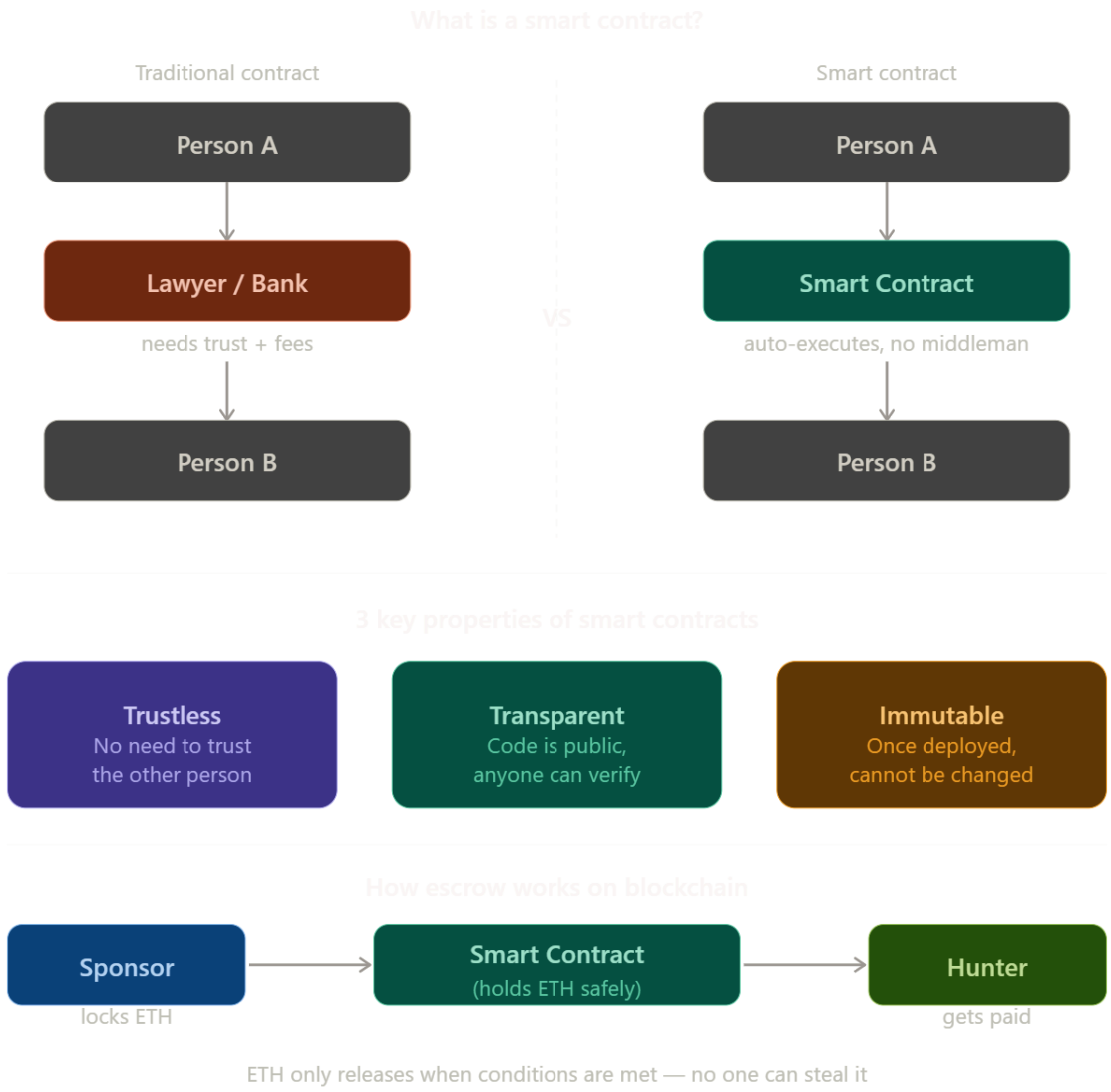
Aim: Creating Smart Contracts and performing transactions using Solidity and Remix IDE

Theory:- Smart contracts are self-executing programs that live on a blockchain. Think of them like a vending machine — you put in money, the machine checks the rules, and automatically gives you what you paid for. No human in the middle.



Decentralized Bug Bounty Escrow System

What is this project?

This is a blockchain-based platform that connects **companies (sponsors)** who want their software tested, with **security researchers (hunters)** who find vulnerabilities — all without trusting a middleman.

When a sponsor posts a bounty and locks ETH, the money is held securely by a smart contract on Ethereum. The hunter only gets paid when their bug report is approved. If there's a disagreement, a neutral arbiter resolves the dispute. Everything is automated, transparent, and tamper-proof.

The Problem it Solves

Traditional bug bounty platforms like HackerOne or Bugcrowd charge high fees, can delay payments for months, and hunters have to trust the company will actually pay. With this system, the ETH is locked upfront — the hunter knows the money is there before they even start working.

The 3 Smart Contracts

BugBountyToken (BBT) — A reward token given to hunters every time they get paid. Acts like a reputation badge — the more bugs you find, the more BBT you earn.

BugBountyEscrow — The heart of the system. It locks ETH, manages the full bounty lifecycle (create → fund → submit → approve/dispute), and automatically pays the hunter when approved.

BugBountyRegistry — A public leaderboard and directory. Stores bounty listings, severity tags, and tracks each hunter's reputation score on-chain for everyone to see.

Who uses it?

Role	What they do
🏢 Sponsor	Posts a bounty, locks ETH, reviews reports
🕵️ Hunter	Finds bugs, submits reports, earns ETH + BBT tokens
⚖️ Arbiter	Resolves disputes fairly between sponsor and hunter
👑 Platform Owner	Collects a small fee, manages the system

Tech Stack

Built with **Solidity 0.8.19** on the **Ethereum** blockchain. Deployable on any EVM-compatible network (Ethereum, Polygon, Base, etc.) and fully testable on **Remix IDE** with no real money needed.

Implementation:-

Step 1:- Deploy Contract 1: BugBountyToken
Step 2:- Deploy Contract 2: BugBountyEscrow

In the **Contract** dropdown, select `BugBountyEscrow`
Fill in the constructor arguments:

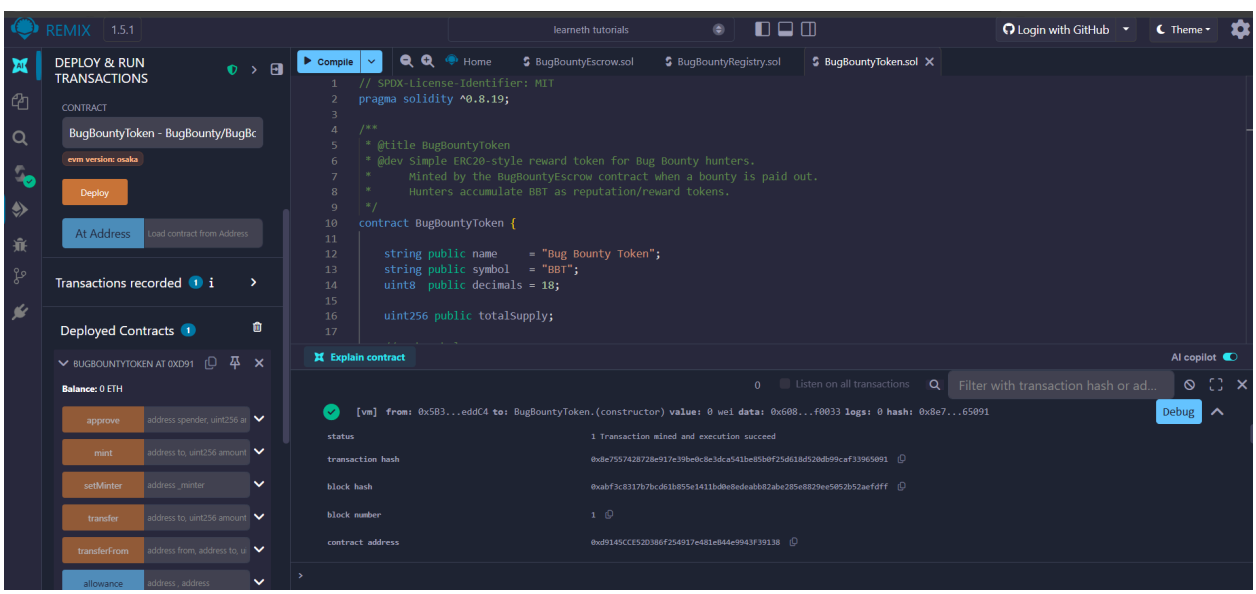
- `_rewardToken`: paste the `BugBountyToken` address from Step 3
- `_platformFeeBps`: enter `200` (= 2% platform fee)

Click **Deploy**
Copy the `BugBountyEscrow` deployed address

Step 3:- Deploy Contract 3: BugBountyRegistry
In the **Contract** dropdown, select `BugBountyRegistry`
Constructor argument:

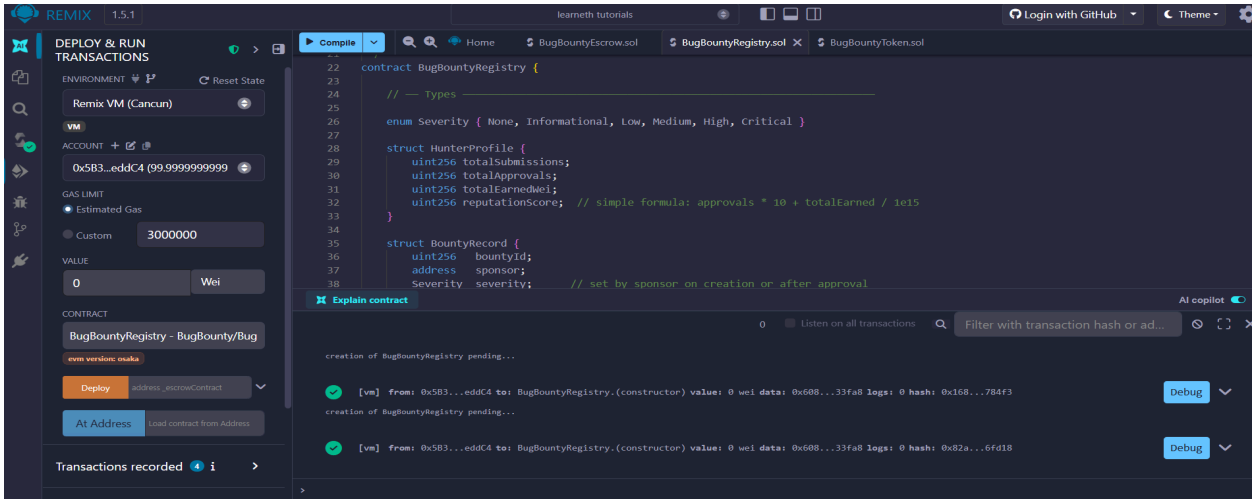
- `_escrowContract`: paste the `BugBountyEscrow` address from Step 4

Click **Deploy**
Copy the `BugBountyRegistry` deployed address

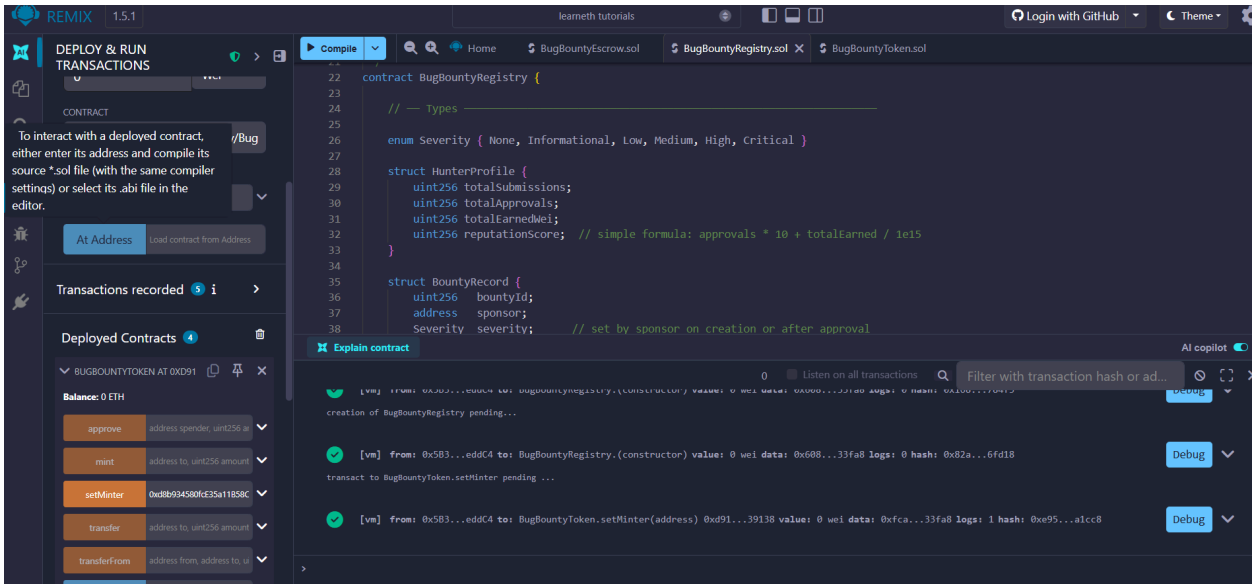


Step 4:- Paste the BugBountyEscrow address to mint BBT tokens then click the transact button

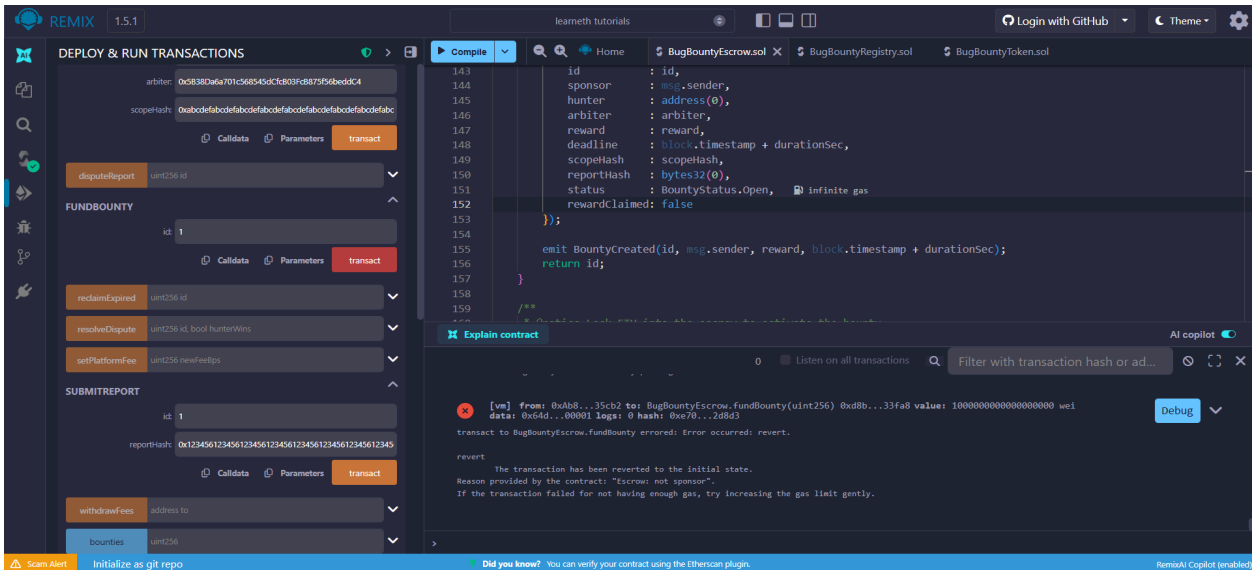
- Expand the deployed `BugBountyToken` contract in the left panel
- Find the `setMinter` function
- Paste the `BugBountyEscrow` address
- Click **transact**



Step 5:- Continuing from the step 4



Step 6:- Call fundBountywith id = 1 and set Value to 1 ETH



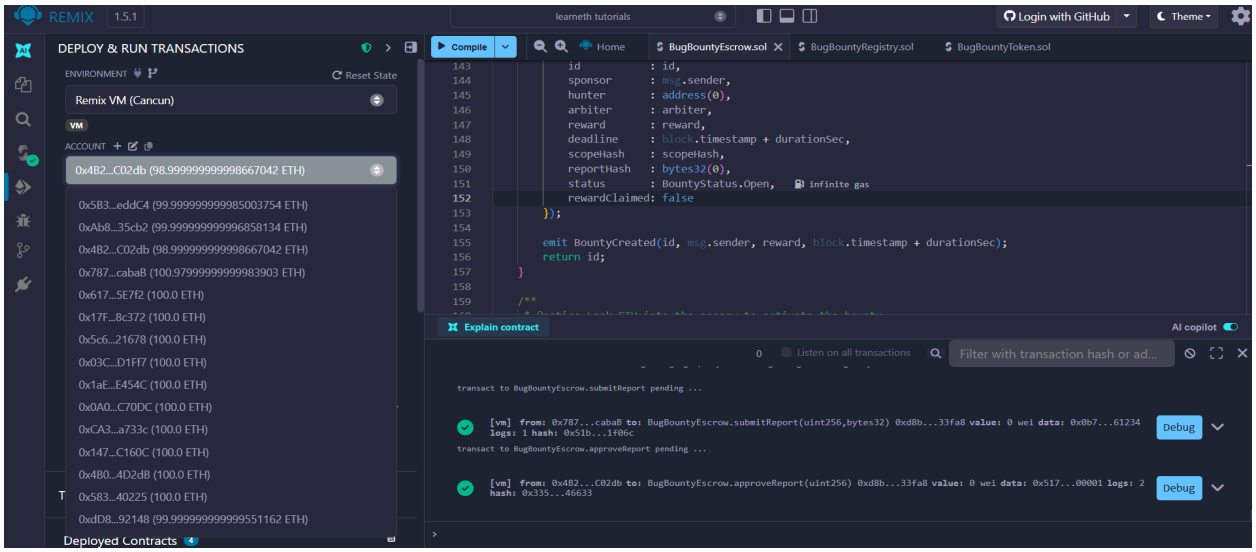
Step 7 :- You need 3 fake people (accounts). At the **top of the Deploy panel** in Remix, there's a dropdown that says **"Account"** Click it — you'll see **10 fake accounts**, each has **100 ETH** (free test money!)

Step 8:- Sponsor creates a bounty

Switch to Account 2 (the Sponsor) from the Account dropdown. Now go to your deployed **BugBountyEscrow** contract and click **`createBounty`**

Fill in that exactly like this
reward → 1000000000000000000
durationSec → 2592000
arbiter → [paste Account 1 address]
scopeHash → 0xabcd….

Step 9:- Register bounty in the registry, then there are 3 people in the accounts part there are 10 fake accounts, then one owner at the top, then there is sponsor, the sponsor will create bounty, sponsor then locks the bounty, hunter submits their bug report and the sponsor approves and then the hunter gets paid, **then the value of that hunter raises**



Conclusion

The Decentralized Bug Bounty Escrow System demonstrates how blockchain technology can solve real trust problems in the cybersecurity industry. By replacing middlemen with smart contracts, it ensures that hunters always get paid fairly and sponsors only release funds when satisfied — all enforced automatically by code, not promises.

With three interconnected contracts handling payments, reputation, and dispute resolution, this system creates a transparent and tamper-proof environment where security researchers can work with confidence. The addition of BBT reward tokens further incentivizes hunters by building their on-chain reputation with every successful submission.

This project proves that decentralized systems are not just a concept — they are practical, deployable tools that can reshape how industries handle trust, payments, and accountability. As Web3 continues to grow, platforms like this represent the future of fair and transparent collaboration between companies and the security community.